DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject : Computer Networks

Class: B.E (CSE/AI & ML) | **Max Marks: 30**
Semester: VI-Sem | **Duration: 1 1/2**
Academic Year: 2025-2026 | **Date : 05-05-2026**

	PART A Answer All Questions (5 x 2 = 10 M)	Marks	BT	CO
1. a)	Write a short note on **piggybacking**. Illustrate the process with a suitable **timing diagram**, and mention one limitation of piggybacking.	2	2	1
b)	Write Stop and Wait protocol efficiency calculation?	2	3	1
c)	Explain Distance Vector Routing Algorithm	2	3	2
d)	Explain Count to Infinity problem and how do you overcome it.	2	2	1
e)	A company has multiple LANs with **250 hosts in each LAN**. If it uses a **Class B** address, determine the **subnet mask in dotted decimal notation**. Also find the **number of subnet bits borrowed** and the **maximum number of subnets possible**.	2	3	1

	PART B Answer Any Two Questions (2 x 10 = 20 M)			
2.	A sender wants to transmit the data bits **1101011011** using CRC. The generator polynomial is **(x^3 + x + 1)** (binary 1011). Determine the CRC **remainder**, the **transmitted codeword**, and explain whether an error is detected if the received bit sequence is **1101011011110**.	5	3	2
	b) Explain in detail the derivation of Ethernet efficiency calculation	5		
3.	a) **Differentiate between Go-Back-N ARQ and Selective Repeat ARQ protocols.**	5	3	2
	b) Explain IPv4 or IP v6 header format	5	4	2
4 a)	An IPv4 datagram of size 4000 bytes is transmitted over a network with an MTU of 1500 bytes. Explain how fragmentation will occur, and determine the number of fragments generated. Consider IP header is 20 Bytes.	5	3	2

b)	Perform **CIDR aggregation** on the following IP address blocks (all with a **/22 prefix**): • 192.168.16.0/22 • 192.168.20.0/22 • 192.168.24.0/22 • 192.168.28.0/22 Determine the **aggregated CIDR block** that represents all of the above networks. Also state the **total number of IP addresses** covered by the aggregated block	5	2	2